

1.3 Computer networks, connections and protocols

Why this unit is taught the way it is

What this unit teaches, and where it ends

15 lessons: 13 with an online experience, 2 paper assessments and 0 bonus challenges. 206 activities in all, worth 338 marks.

By the end of it a pupil should be able to:

- know what is meant by a standalone computer nw-101
- know the different types of network: LAN and WAN nw-101
- understand the advantages of networking nw-101
- understand the implications (disadvantages) of networking nw-101
- understand what factors can affect the performance of a network nw-102
- be able to explain how each factor affects network speed nw-102
- know what a client-server model is nw-103
- know what a peer-to-peer model is nw-103
- understand the different roles computers have in each model nw-103
- know the hardware needed to connect a LAN nw-104
- understand the purpose of each piece of hardware and the jobs it does nw-104
- understand what the internet actually is nw-105
- understand the terms Domain Name Service and domain name server nw-105
- understand what is meant by hosting and the cloud nw-105
- understand what is meant by the terms web server and client nw-105
- know what a star network is nw-106
- know what a mesh network is nw-106
- understand that the internet is an example of a partial mesh network nw-106
- know the advantages and disadvantages of star and mesh networks nw-106
- know which lessons I am secure on and which I need to revise nw-107
- have completed any outstanding work from lessons 1 to 6 nw-107
- be ready for my mid-topic test next lesson nw-107
- understand that Ethernet is a wired method of connection nw-109
- understand that Wi-Fi and Bluetooth are wireless methods of connection nw-109
- understand the benefits and drawbacks of wired versus wireless connections nw-109
- be able to recommend a connection type for a specified scenario nw-109
- know the basics of how cryptography can work with a simple key nw-110
- know how wireless devices authenticate each other before communicating data nw-110
- understand the difference between private and public keys nw-110
- understand why private keys are never shared nw-110
- understand the uses of MAC and IP addressing nw-111
- understand the difference between IPv4 and IPv6 nw-111
- understand the need for IPv6 nw-111

- understand the need for standards in computing nw-112
- understand the seven common protocols and what they are used for nw-112
- know why protocols are layered nw-113
- understand the advantages of layering nw-113
- know which lessons I am secure on and which to revise nw-114
- be ready for the end-of-unit assessment nw-114

Why the lessons are in this order

Each lesson's starter retrieves the one before it, so the order is the retrieval schedule as much as the teaching one. The chain for this unit:

LESSON	TEACHES	RETRIEVES
1 Types of network: zoom out	know what is meant by a standalone computer	nothing before it in this unit
2 Network control room	understand what factors can affect the performance of a network	Types of network: zoom out
3 Network showdown	know what a client-server model is	Network control room
4 Mission: wire up the school	know the hardware needed to connect a LAN	Network showdown
5 Inside the internet	understand what the internet actually is	Mission: wire up the school
6 Star and mesh networks	know what a star network is	Inside the internet
7 Revision HQ	know which lessons I am secure on and which I need to revise	Star and mesh networks
8 Assessment 1	assesses the whole unit	Revision HQ
9 Connection café	understand that Ethernet is a wired method of connection	Assessment 1
10 Cipher room	know the basics of how cryptography can work with a simple key	Connection café
11 Address sorting office	understand the uses of MAC and IP addressing	Cipher room
12 Protocol city	understand the need for standards in computing	Address sorting office
13 Parcel depot	know why protocols are layered	Protocol city
14 Assessment 2 reflection	assesses the whole unit	Parcel depot
14 Revision HQ 2	know which lessons I am secure on and which to revise	Assessment 2 reflection

The assessments in this unit are placed where it is so that the reflection sheet still has lessons left to send a pupil back to.

What pupils find hard here

None of this unit's walls names a common mistake in so many words. The misconceptions it works against are in the wrong options of its questions instead: every one of those is a plausible wrong idea rather than filler, and the lesson plans list them question by question.

Every multiple-choice and Predict activity in the unit — 180 of them — offers wrong answers that are the misconception, not filler. The split across those options at the hinge question is the cheapest diagnostic in the lesson.

Where the explaining, modelling and practising happen

PHASE	WHERE IT IS
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Retrieve	The starter on the worksheet, before any device is opened. It is answered on paper and corrected before the lesson moves on.
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Explain	The slides and the station walls carry the same explanation, so a teacher can teach from the front and a pupil can re-read it in the room.
Model	There is no worked example on screen in this unit: its activities are questions to answer rather than techniques to copy. The modelling is yours, from the slides, before anyone starts - and it is the part to protect when the lesson runs short.
Practise	189 guided activities, then 17 independent ones.
Check	The badge questions mark as the pupil goes; the hinge question is for the class at once.
Apply	0 challenge activities, and the exam practice on the worksheet.
Review	The exit question and the confidence columns. The confidence columns supplement the evidence; they do not stand in for it.

How the website, the slides and the worksheet divide the work

They are not three copies of the same thing.

- **The slides** are for the front of the room: the explanation and the model, before anyone is working alone.
- **The experience** is where a pupil reads, decides and is marked. It gives the immediate feedback a teacher cannot give thirty pupils at once.
- **The worksheet** is the record and the writing. The key fact in their own words, the station challenge, the exam practice. It is also what survives when the devices do not.

The rule that holds them together is **write first, then answer at the badge**. A pupil who taps first has turned the lesson into a quiz game, and the worksheet is how you can see at a glance who did.

Why a 360° room, and what it should produce

The room is not there to be impressive. It is there because this unit's content is a set of related parts that a pupil has to hold together, and putting one part on each wall makes the set visible and navigable: a pupil can turn back to station 2 while answering station 4. The evidence it should produce is on paper — 42 key facts and 71 station challenges in a pupil's own words — and in the score screen, which shows which station a pupil lost marks on rather than one number.

No claim is made here that immersion improves learning on its own. What the format does is make the structure of a topic physical and keep a pupil's attention on one part at a time; the teaching is in the explanation, the example and the practice, as it would be anywhere else.

Support, and how it is taken away

- The worked example is on the early activities of a station and not on the late ones. That fade is deliberate.
- The hint is a ladder asked for one rung at a time, and the top of it is still the technique on different data, never the answer.
- Read aloud is on every task.
- Prescribed values are boxed and in a monospace face as well as coloured, so they survive a black-and-white printout and a high-contrast mode.
- **Lower literacy and EAL:** pre-teach the vocabulary (Standalone computer, LAN, WAN, Bandwidth, Latency, Transmission media, Error rate, Client-server, Peer-to-peer, Switch, Router, Network interface card). Accept the key fact spoken before it is written, and in two or three words.
- **SEND:** pace is not fixed. Progress saves after every answer and reopens at the earliest unfinished question, so a pupil may finish a station next lesson without losing anything.
- A headset is never required.

Checking, and what to do with what you find

Three checks, in this order: the starter (did last lesson stick), the hinge question (is the class ready for independent work), the exit question (did this lesson stick). The first two change what you do in the lesson; the third changes what you start the next one with.

When a hinge question splits the class, go back to the station it sits on and work its example again, aloud, before anyone goes on. When a test comes back, the reflection sheet names the topics to revisit and review mode on those lessons reopens exactly the questions a pupil got wrong.

Where the approach comes from

The shape of a lesson here — retrieve, explain, model, practise with support that fades, check, apply, review — is ordinary evidence-informed practice rather than anything this product invented. Useful starting points, none of them about Revise 360:

- Rosenshine, *Principles of Instruction* (American Educator, 2012) — the daily review, small steps, worked examples and checking for understanding that this structure follows.
- Sweller, van Merriënboer and Paas on cognitive load and the worked-example effect — why the example comes before the independent task and why the support is withdrawn gradually.
- Dunlosky et al., *Improving Students' Learning With Effective Learning Techniques* (2013) — the evidence for retrieval practice and spacing, which the starters and the review mode use.
- Wiliam, *Embedded Formative Assessment* — hinge questions and what to do with the answers.

What is not claimed. Nothing above is evidence that Revise 360 improves outcomes. No trial of this product has been run, and immersion is not claimed to improve learning by itself. The citations are for the teaching structure; whether these resources work in your classroom is a question your own assessment answers.

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